JAVA Object-Oriented Programming

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Section: D.
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Exercise Title: Java OOP – Cafe Ordering System

I. CONCEPT EXPLANATION

In this activity, we developed a Café Ordering System using Java and Object-Oriented Programming (OOP) concepts. The program allows customers to order different menu items, calculate prices, apply discounts and vouchers, print receipts, and earn loyalty points.

The following OOP concepts were applied in our program:

1. **Encapsulation** - We used encapsulation to protect the data in our classes using private and protected attributes with getter and setter methods. This helps prevent invalid values and keeps the data organized.

2. **Inheritance** - We used inheritance when the Beverage, Pastry, and Meal classes extended the MenuItem class. This allows them to reuse common properties and methods.

3. **Abstraction** - We used abstraction by making MenuItem an abstract class with an abstract computePrice() method. This allows each menu item to calculate its price differently.

4. **Polymorphism** - We used polymorphism by overriding the computePrice() and getDescription() methods in the Beverage, Pastry, and Meal classes. This allows each class to perform the same method in its own way.

5. **Constructor Overloading** - We used constructor overloading in the MenuItem and Customer classes to create objects using different sets of parameters. This makes object creation more flexible.

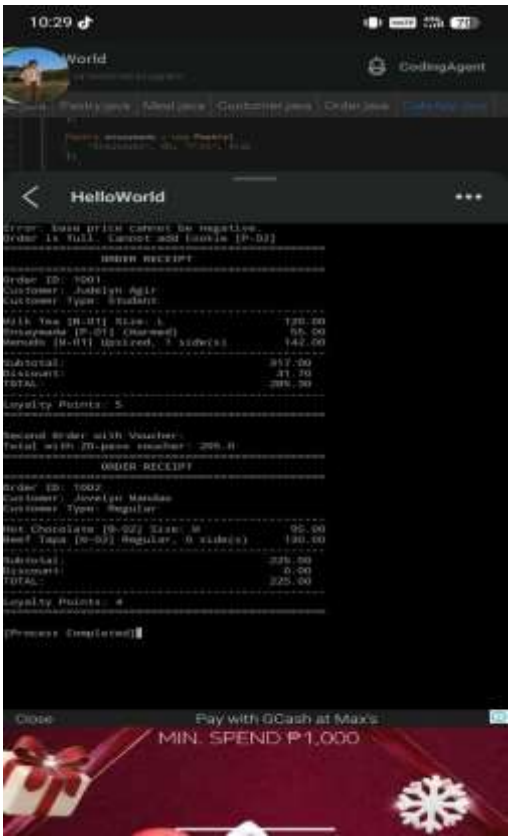
6. **Method Overloading** - We used method overloading in the Order and Customer classes through methods with the same name but different parameters. This allows the program to perform similar operations in different ways.

II. PROGRAM OUTPUT

The program demonstrates how the Café Ordering System works by creating two customers and different menu items, including beverages, pastries, and meals.

It also demonstrates adding menu items to an order, calculating the subtotal, applying a student discount and voucher, and displaying the final receipt with the customer’s loyalty points.

Attached below are the screenshot of the actual program output :



III. CHALLENGES ENCOUNTERED AND SOLUTIONS

During the development of our program, we encountered several challenges, such as repeated coding errors, difficulties connecting the classes properly, and problems with matching methods and parameters. We addressed these issues by carefully reviewing our codes, correcting mistakes, checking the connections between classes, and testing the program repeatedly until everything worked prproperlyl

IV. CONCLUSION

Overall, this activity helped us understand and apply the different concepts of Object-Oriented Programming in Java. Throughout the development of our Café Ordering System, we learned how to create and connect different classes, use methods, calculate prices, and fix coding errors. Although we encountered several challenges, we worked through them by checking and improving our codes. This experience helped us improve our programming skills, teamwork, and understanding of how Java OOP works in creating a functional program.